

Task 3

Exploratory Data Analysis (EDA)

Tools Used

Python, Pandas, Matplotlib, Seaborn, Jupyter Notebook

Introduction

This project focuses on performing Exploratory Data Analysis (EDA) on a dataset such as the Iris Dataset or Tips Dataset. The analysis helps understand data distributions, relationships between variables, correlations, patterns, trends, and anomalies. EDA is an important step before building machine learning models because it provides insights into the dataset and improves decision-making.

Objective

- Perform Univariate Analysis
- Perform Bivariate Analysis
- Generate Correlation Heatmaps
- Create Pairplots
- Identify patterns, trends, and outliers
- Summarize key insights from the dataset

Tools & Technologies Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook

Dataset Used

Dataset Name: Iris Dataset / Tips Dataset

The dataset contains multiple numerical and categorical variables used to analyze distributions, relationships, and correlations.

Steps Performed

1. Imported required libraries
2. Loaded the dataset
3. Explored dataset structure
4. Checked missing values
5. Performed Univariate Analysis using Histograms and Box Plots
6. Performed Bivariate Analysis using Scatter Plots and Bar Charts

7. Generated Correlation Matrix and Heatmap
8. Created Pairplots for numerical features
9. Identified trends, patterns, and anomalies
10. Summarized findings and insights

Important Code Snippets

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('dataset.csv')
df.head()
df.info()
df.describe()
sns.heatmap(df.corr(), annot=True)
sns.pairplot(df)
```

Univariate Analysis

- Histograms used to understand feature distributions
- Box Plots used to identify outliers
- Summary statistics generated using describe()

Bivariate Analysis

- Scatter Plots used to study relationships between variables
- Bar Charts used for comparison of categories
- Correlation values used to measure feature relationships

Correlation Analysis

A correlation heatmap was generated to visualize the strength and direction of relationships between numerical variables.

Key Observations

- Some variables showed strong positive correlations
- Certain features contained outliers
- Pairplots helped visualize class separation and relationships
- Data distributions revealed important patterns

Learning Outcome

- Learned how to perform Exploratory Data Analysis
- Understood data distributions and relationships
- Learned to identify outliers and trends visually
- Improved data interpretation skills using visualizations

Conclusion

This task provided practical experience in analyzing datasets using EDA techniques. The visualizations and statistical summaries helped uncover meaningful insights and prepare the data for further machine learning tasks.